

PP Lecture 3 PDC

Part 1: Past Papers

QUIZ 01-B SPRING 2023 / QUIZ 01 FALL 2022 • QUESTION 1

Q: What value may assist us in finding out if parallel overhead exists when running a program using multiple processors in parallel?

- a. Karp-Flatt Metric
- b. Theoretical Speedup (Amdahl's Law)
- c. Moore's Law
- d. Fraction of the program that is parallelizable

A: a. Karp-Flatt Metric

ASSIGNMENT SPRING 2023 / QUIZ 01-B SPRING 2023 / SESSIONAL-I SPRING 2023 • QUESTION 2

Q: Suppose we have a system with one processor and a serial program. We want to upgrade this system and we have the following two options:

- i. Increase the number of processors from 1 to 4 and parallelize 70% of the code.
- ii. Increase the number of processors from 1 to 8 and parallelize 40% of the code.

Calculate speedup in each case and identify which one is better?

A:

Option i (4 processors, 70% parallelized):

$$\text{Speedup} = 1 / ((1 - 0.70) + 0.70/4) = 1 / (0.30 + 0.175) = 1 / 0.475 \approx 2.10$$

Option ii (8 processors, 40% parallelized):

$$\text{Speedup} = 1 / ((1 - 0.40) + 0.40/8) = 1 / (0.60 + 0.05) = 1 / 0.65 \approx 1.54$$

Conclusion: Option i (parallelizing 70% of the code with 4 processors) is better as it yields a higher speedup (2.10 vs 1.54).

MIDTERM 1 SPRING 2023 / SESSIONAL-I SPRING 2023 • QUESTION 3

Q: 70% of a program's execution time occurs inside a loop that can be executed in parallel and rest 30% in serial. What is the maximum speedup we should expect from a parallel version of the program executing on 8 CPUs?

- a. 2.58
- b. 3.06
- c. 3.50
- d. 4.08

A: a. 2.58

Calculation: $\text{Speedup} = 1 / (0.30 + 0.70/8) = 1 / 0.3875 \approx 2.58$

MIDTERM 1 SPRING 2023 / SESSIONAL-I FALL 2024 • QUESTION 4

Q: Assume a sequential program S has an execution time of 500 seconds. Further, assume that Sp is a parallel variant of S. After an experimental evaluation over different number of processors, the following running times were achieved:

Processors (P): 2, 4, 6, 8

Execution Time (seconds): 294, 203.84, 176.62, 164.92

a) Calculate Speedups for each of the experimental configurations.

b) Calculate the Karp-Flatt metric values. Interpret the results of Karp-Flatt metric and write your opinion accordingly.

A:

a) Speedups ($S = T_1 / TP$):

$$P = 2: 500 / 294 = 1.70$$

$$P = 4: 500 / 203.84 = 2.45$$

$$P = 6: 500 / 176.62 = 2.83$$

$$P = 8: 500 / 164.92 = 3.03$$

b) Karp-Flatt Metric ($e = (1/S - 1/P) / (1 - 1/P)$):

$$P = 2: ((1/1.70) - (1/2)) / (1 - (1/2)) = 0.18$$

$$P = 4: ((1/2.45) - (1/4)) / (1 - (1/4)) = 0.21$$

$$P = 6: ((1/2.83) - (1/6)) / (1 - (1/6)) = 0.22$$

$$P = 8: ((1/3.03) - (1/8)) / (1 - (1/8)) = 0.23$$

Interpretation: Since e is steadily increasing with P, it suggests that parallelization overhead is also contributing to poor speedup. Hence, we need to reduce this overhead to improve speedups.

MIDTERM I SPRING 2022 / SESSIONAL-I SPRING 2024 • QUESTION 5

Q: Which one is true about Amdahl's law formulation:

- A) It does not account for communication overheads
- B) It provides an upper-bound on the achievable speedups
- C) If the number of processors approaches infinity, Speedup is bounded by 1 / sequential fraction
- D) All (a), (b), and (c) are correct
- E) Only (b) and (c) are correct

A: D) All (a), (b), and (c) are correct

MIDTERM I SPRING 2022 • QUESTION 6

Q: Assume a sequential program S has an execution time of 200 seconds. Further, assume Sp is a parallel variant of S. After an experimental evaluation over different number of processors, the following running times were achieved:

Processors (P): 2, 4, 6, 8

Execution Time (seconds): 107, 61.92, 48.31, 42.46

- a) Calculate Speedups for each of the experimental configurations.
- b) Calculate the Karp-Flatt metric values and interpret the results.

A:

a) Speedups:

$$P = 2: 200 / 107 = 1.87$$

$$P = 4: 200 / 61.92 = 3.23$$

$$P = 6: 200 / 48.31 = 4.14$$

$$P = 8: 200 / 42.46 = 4.71$$

b) Karp-Flatt Metric:

$$P = 2: 0.07 \mid P = 4: 0.08 \mid P = 6: 0.09 \mid P = 8: 0.10$$

Interpretation: As e is steadily increasing with P, it suggests that parallelization overhead is contributing to poor speedup. Reducing this overhead is necessary to improve speedup.

SESSIONAL-I EXAM SPRING 2024 • QUESTION 7

Q: Karp-Flatt metric can give us information for which of the following?

- A) Serial code in terms of number of lines-of-code
- B) Time proportion spent executing serial parts of the code
- C) Any parallelization overheads
- D) Number of processors used for parallelization

A: (B) and (C)

Q: Consider the following equation in the context of Karp-Flatt metric:

$$f = (1/s - 1/p) / (1 - 1/p)$$

- (a) If p is the number of processors, what is s ?
- (b) What is f ?
- (c) What are the units of f ?
- (d) Following data was gathered experimentally on some parallel computer A:

p : 1, 2, 3, 4, 8

Time (s): 2.17, 1.11, 0.754, 0.577, 0.312

s : ---, 1.95, 2.88, 3.76, 6.96

f : ---, 0.024, 0.021, 0.021, 0.021

What can we infer about the quality of parallelism by looking at the data provided? Please point to specific data to support your answer.

A:

- (a) s is the speed-up achieved.
- (b) f is the Karp-Flatt metric (experimentally determined serial fraction).
- (c) It is a unitless entity as it represents a time fraction.
- (d) **Inference:** (1) Speed-up increases consistently as processors increase. (2) The Karp-Flatt metric value remains almost constant across runs ($f \approx 0.021$). This implies parallelization overheads are negligible, but there is a limited amount of inherent parallelism available in the problem (approx. 2.1% is inherently sequential).

Q: How does Gustafson's law differ from Amdahl's law?

A: In Gustafson's Law, we increase the input dataset / workload size in proportion to the increasing number of processors. In Amdahl's law, we keep the problem size fixed across runs as processors increase.

Q: Derive Gustafson's law using the symbols (s = serial fraction, p = parallel fraction, N = processors). Clearly mention what each symbol means. Your final expression should be in terms of N and s only. What are the two assumptions made in this derivation?

A:

Symbols: s = serial proportion of time, p = parallel proportion of time (where $s + p = 1$), N = number of processors.

Derivation:

Execution time on N processors = $s + p = 1$.

Hypothetical execution time on a single serial processor for the scaled workload = $s + N \times p$.

$$\text{Scaled Speedup} = (s + p \times N) / (s + p) = s + p \times N$$

Substituting $p = 1 - s$:

$$\text{Scaled Speedup} = s + (1 - s) \times N = N + (1 - N) \times s$$

Assumptions:

1. Each processor is equally capable and divides/finishes allocated work at the same rate.
2. There are no communication or synchronization overheads, and workload scaling continues perfectly.

SESSIONAL-I EXAM SPRING 2025 • QUESTION 11

Q: Assume that 90% of an algorithm can be parallelized while 10% must be executed sequentially (s=0.1, p=0.9).

- Calculate the missing values (A, B, C, D) in the table (provided in original prompt).**
- Review K-F metric derived from Amdahl's speedup. What conclusions can you draw, and why?**
- Review K-F metric derived from Gustafson's speedup. What conclusions can you draw, and why?**
- For a fixed workload, would you expect experimental K-F values to decrease, increase, or remain constant, and why?**

A:

a) Calculated values:

A (Amdahl P=2): $1 / (0.1 + 0.9/2) = 1.818$

B (Gustafson P=2): $0.1 + 0.9 \times 2 = 1.9$

C (K-F Amdahl P=2): $((1/1.818) - 0.5) / (1 - 0.5) = 0.1$

D (K-F Gustafson P=2): $((1/1.9) - 0.5) / (1 - 0.5) = 0.053$

b) Amdahl K-F Conclusion: The serial fraction remains constant (0.1) because problem size is fixed and theoretical Amdahl speedup assumes zero parallel overheads.

c) Gustafson K-F Conclusion: The serial fraction decreases (0.053 → 0.014) because as problem size grows with P, the parallelizable workload dominates.

d) Experimental K-F Trend: Experimental K-F values will increase as P increases because empirical execution incurs real-world parallelization overheads (communication, synchronization) ignored in theoretical models.

SESSIONAL-I EXAM SPRING 2025 • QUESTION 12

Q: You're optimizing a scientific computation program where profiling reveals that 70% of the work is parallelizable while 30% remains sequential. You plan to rent a server from a cloud provider offering machines with 2 to 500 cores. You prefer speed-up factors expressed as whole numbers (e.g., 3x, 4x) rather than fractions. Which core count should you choose?

A: 21 cores

Explanation: By Amdahl's Law, maximum speedup with infinite cores is $1 / 0.30 \approx 3.33$. The maximum achievable whole-number speedup is 3x.

Setting Speedup = 3 in Amdahl's formula:

$$3 = 1 / (0.30 + 0.70/N) \Rightarrow 0.30 + 0.70/N = 1/3 \Rightarrow 0.70/N = 1/30 \Rightarrow N = 21 \text{ cores}$$

FINAL EXAM SPRING 2024 • QUESTION 13

Q:

- (a) Write the Amdahl's law. Clearly mention what each symbol represents and their units.
- (b) If a program needs 20 hours to complete using a single processor, but a 1-hour portion cannot be parallelized, what is the minimum possible execution time?
- (c) Assume a program is split into four consecutive parts with execution time percentages $p_1 = 0.11$, $p_2 = 0.18$, $p_3 = 0.23$, $p_4 = 0.48$. Part 1 is not sped up, Part 2 is sped up 5 times, Part 3 is sped up 20 times, and Part 4 is sped up 1.6 times. What is the overall speedup?

A:

- (a) **Amdahl's Law:** $\text{Speedup} = 1 / (F + (1-F)/P)$. Symbols: F = serial fraction (dimensionless ratio), $1-F$ = parallel fraction (dimensionless ratio), P = number of processors (dimensionless count), Speedup = performance ratio (dimensionless). Both sides are unitless.
- (b) **Minimum possible execution time** = 1 hour (when $P \rightarrow \infty$, the 19-hour parallel part approaches 0 hours, leaving only the 1-hour sequential part).
- (c) **Overall Speedup:**

$$\text{New Execution Time} = 0.11 + 0.18/5 + 0.23/20 + 0.48/1.6 = 0.11 + 0.036 + 0.0115 + 0.30 = 0.4575$$

$$\text{Overall Speedup} = 1 / 0.4575 \approx 2.186 \text{ (or 2.19)}$$

FINAL EXAM SPRING 2023 - PAPER.PDF • QUESTION 14

Q: Assume a sequential program S takes 400 seconds. A parallel variant takes 85.55 seconds on 8 processors. What is the Karp-Flatt metric?

- a. 1.15
- b. 0.129
- c. 0.99
- d. 0.007

A: d. 0.007

FINAL EXAM SPRING 2023 - PAPER.PDF • QUESTION 15

Q: Assume a sequential program S takes 400 seconds. A parallel variant takes 55.55 seconds on 8 processors. What is the approximate speedup?

- a. 50
- b. 7.2
- c. 0.9
- d. 57.6

A: b. 7.2 ($400 / 55.55 \approx 7.20$)

Q: Explain Amdahl's law and its shortcomings.

A: Amdahl's Law states that maximum speedup is bounded by $1 / (F + (1-F)/P)$, where F is the sequential fraction.

Shortcomings:

1. Ignores parallel overheads: Does not account for communication latency, synchronization, or thread management.
2. Assumes fixed problem size: Does not reflect real-world scaling where users run larger problems as more computing power becomes available.

Q: In Karp-Flatt metric, if e increases as p increases, it means that parallelization overhead is affecting the speedup.

- a. True
- b. False

A: a. True

Q: What is Amdahl's law? According to this law, if we want to double the overall speed of a system having 60% parallelizable code, then how many processors would be required?

A:

Parallelizable code $P = 0.60 \Rightarrow$ Serial fraction $1-P = 0.40$.
Target Speedup = 2.

$$2 = 1 / (0.40 + 0.60/N) \Rightarrow 0.40 + 0.60/N = 0.50 \Rightarrow 0.60/N = 0.10 \Rightarrow N = 6 \text{ processors}$$

Q: Given execution results for System A (4 cores, 80% efficiency, 10s execution time) and System B (8 cores, 60% efficiency, 6s execution time):

- a. Calculate the speedup of System B over System A.
- b. Which system is more efficient per core?

A:

a. Speedup of B over A: $\text{Time}_A / \text{Time}_B = 10 / 6 \approx 1.67x$

b. Per-core efficiency:

System A: $80\% / 4 = 20\% \text{ per core}$

System B: $60\% / 8 = 7.5\% \text{ per core}$

System A is more efficient per core.

LECTURE 3 SLIDES EXAMPLES • QUESTION 20

Q: Benchmarking a parallel program on 2 to 8 processors produces speedups:

Case 1 (S): 1.82, 2.50, 3.08, 3.57, 4.00, 4.38, 4.71 (e = 0.10 constant)

Case 2 (S): 1.87, 2.61, 3.23, 3.73, 4.14, 4.46, 4.71 (e increases from 0.07 to 0.10)

Identify the primary reason for poor speedup in each case.

A:

Case 1: Inherent sequential fraction (10%). Since e is constant, parallel overhead is negligible.

Case 2: Parallelization overhead. Since e increases steadily with P, overhead dominates speedup loss.

Part 2: Book Questions (Introduction to Parallel Computing)

CHAPTER 5 - PROBLEM 5.1 • QUESTION 1

Q: (Amdahl's law [Amd67]) If a problem of size W has a serial component W_S , prove that W/W_S is an upper bound on its speedup, no matter how many processing elements are used.

A:

Let total work $W = W_S + W_P$, where W_S is serial work and W_P is parallelizable work.

The parallel execution time on p processing elements satisfies:

$$T_P \geq W_S + W_P / p$$

The speedup S is given by:

$$S = T_S / T_P = W / T_P \leq W / (W_S + W_P / p)$$

Taking the limit as $p \rightarrow \infty$:

$$\lim(p \rightarrow \infty) S \leq W / (W_S + 0) = W / W_S$$

Since $W_P / p \geq 0$ for any finite number of processors, $T_P \geq W_S$, proving that W / W_S is an upper bound on speedup.

CHAPTER 5 - PROBLEM 5.5 • QUESTION 2

Q: (Scaled speedup) Scaled speedup is defined as the speedup obtained when the problem size is increased linearly with the number of processing elements; that is, if W is chosen as a base problem size for a single processing element, then $\text{Scaled speedup} = (p W) / T_P(p W, p)$. For adding n numbers on p processing elements, compare the standard speedup curve for base problem size $n = 256$ with the scaled speedup curve for $p = 1, 4, 16, 64, 256$ (communication time per number = 10, addition time = 1).

A:

Parallel execution time formula for adding n numbers: $T_P = (n / p - 1) + 11 \log_2(p)$.

Standard Speedup (fixed $n = 256$):

$$S_{\text{standard}}(p) = 256 / (256/p - 1 + 11 \log_2(p))$$

As p grows, $256/p \rightarrow 1$, and speedup saturates rapidly due to Amdahl's law.

Scaled Speedup (problem size scales linearly $n = 256 p$):

$$T_P(256 p, p) = (256 - 1) + 11 \log_2(p) = 255 + 11 \log_2(p)$$

$$\text{Scaled speedup} = 256 p / (255 + 11 \log_2(p))$$

Unlike standard speedup, scaled speedup increases nearly linearly with p because computation per processor remains constant (256) while communication grows only logarithmically ($11 \log_2(p)$).

CHAPTER 5 - PROBLEM 5.8 • QUESTION 3

Q: A drawback of increasing the number of processing elements without increasing the total workload is that the speedup does not increase linearly with the number of processing elements, and the efficiency drops monotonically. Discuss whether or not scaled speedup increases linearly with the number of processing elements in general. What can you say about the isoefficiency function of a parallel system whose scaled speedup curve matches the speedup curve determined by increasing the problem size according to the isoefficiency function?

A:

Scaled speedup does not generally increase strictly linearly with p unless parallel overhead $T_O = 0$. If overhead $T_O(p W_1, p)$ grows faster than p , efficiency will still decrease.

If the scaled speedup curve matches the speedup curve obtained by scaling according to the isoefficiency function, it implies that the isoefficiency function is linear ($\Theta(p)$), indicating a highly scalable parallel system.

CHAPTER 5 - PROBLEM 5.9 • QUESTION 4

Q: (Time-constrained scaling) Using $T_P = (n/p - 1) + 11 \log_2(p)$ for $p = 1, 4, 16, 64, 256, 1024, 4096$, what is the largest problem that can be solved if total execution time is not to exceed 512 time units? In general, is it possible to solve an arbitrarily large problem in a fixed amount of time provided unlimited processing elements are available? Why?

A:

Setting $T_P \leq 512 \Rightarrow n/p - 1 + 11 \log_2(p) \leq 512 \Rightarrow n \leq p (513 - 11 \log_2(p))$.
For $p = 4096$ ($\log_2(4096) = 12$):

$$n \leq 4096 \times (513 - 11 \times 12) = 4096 \times 381 = 1,560,576 \text{ numbers}$$

In general, NO. It is not possible to solve an arbitrarily large problem in fixed time even with unlimited processors because communication/synchronization overhead T_O (e.g., $11 \log_2(p)$) grows with p and will eventually exceed the time limit.

CHAPTER 5 - PROBLEM 5.12 • QUESTION 5

Q: Prove that if $T_O = \Theta(p)$ for a given problem size, then the parallel execution time will continue to decrease as p is increased and will asymptotically approach a constant value. Also prove that if $T_O > \Theta(p)$, then T_P first decreases and then increases with p ; hence, it has a distinct minimum.

A:

Parallel execution time is $T_P = (W + T_O(W, p)) / p = W/p + T_O(W, p)/p$.

Case 1 ($T_O = c \cdot p$):

$$T_P = W/p + c$$

As $p \rightarrow \infty$, $W/p \rightarrow 0$, so $T_P \rightarrow c$ (a constant). T_P decreases monotonically toward c .

Case 2 ($T_O > \Theta(p)$, e.g., $\Theta(p \log p)$ or $\Theta(p^2)$):

The term $T_O(W, p)/p$ strictly increases with p . For small p , W/p dominates and T_P decreases. For large p , T_O/p dominates and T_P increases. Thus, $T_P(p)$ possesses a distinct global minimum.